

Garvit Patwa

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SUMMARY

B.Tech Computer Science undergraduate with hands-on experience developing full-stack web applications using React.js, Node.js, Express.js, and MongoDB, along with IoT-based systems using ESP32, MQTT, and C++. Solved 200+ Data Structures & Algorithms problems and enjoy building software that solves real-world problems.

SKILLS

Languages: C++, C, Python, JavaScript, SQL

Web Development: HTML, CSS, JavaScript, React.js, Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks

Developer Tools: Git, GitHub, VS Code, Figma

Embedded Systems: Arduino, ESP32, IoT, PZEM-004T

WORK EXPERIENCE

Trading Engineers

On-site Internship, India

UI/UX Design Intern | CCMS (Centralized Control & Monitoring System)

June 2025 - July 2025

- Designed a scroll-free dashboard for monitoring 15,000+ smart streetlights, enabling operators to view faults, complaints, and system status from a single interface.
- Created GIS-based interface designs and role-based workflows, improving navigation and operational efficiency for control-room users.
- Collaborated with cross-functional teams to translate operational requirements into production-ready Figma prototypes for real-time monitoring and automation.

PROJECTS

Hierarchy Management System | Full-Stack Web Application

2026

Technologies: React.js, Node.js, Express.js, MongoDB, JWT, Git, GitHub

- Developed a full-stack Hierarchy Management System using React.js, Node.js, Express.js, and MongoDB to manage multi-level administrative hierarchies.
- Designed REST APIs with JWT-based authentication and role-based authorization to provide secure access for Admin and Operator users.
- Implemented PDF and Excel report generation with Git/GitHub-based version control to streamline reporting and collaborative development.

IoT-Based Smart Street Light Monitoring & Control System (Hardware Project)

2025

Technologies: ESP32, C++, Arduino IDE, MQTT, Quectel EC200U, GPS, IoT, Relay Module

- Developed an IoT-based Smart Street Light Monitoring & Control System using ESP32, MQTT, and the Quectel EC200U 4G module for real-time remote communication.
- Integrated GPS tracking, MQTT publish/subscribe messaging, and relay control to enable live device monitoring and remote streetlight operation.
- Implemented automatic MQTT reconnection and network recovery mechanisms to improve communication reliability in unstable network conditions.

ACHIEVEMENTS

- Secured 4th Place in IDEATHON at Jaypee University, Guna for developing an IoT-based Smart Street Light Monitoring & Control System.
- Solved 200+ Data Structures & Algorithms problems across LeetCode and Codeforces (Codolio Profile).

EDUCATION

JAYPEE UNIVERSITY OF ENGINEERING & TECHNOLOGY

Guna, India

Bachelor of Technology in Computer Science & Engineering

2023-2027

CGPA: 7.1 / 10

CERTIFICATIONS

- Internet of Things Foundation Certification
- Mastering IOT with Arduino

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